4. Loops

ICT360

Tech Flash

Sprite Library : The sprite library allows the user to select a sprite from a wide range of collections. They are split into various categories like animals, food, fantasy, etc.

Backdrop Library : The backdrop library allows the user to select a backdrop from a wide range of collections. They are split into various categories like animals, food, fantasy, etc.

Speech Block : A speech block is used to display dialogues.

Exercise

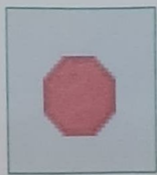
Tick Mark ☑ the correct answer

4 Which block is used to add sound to the animation?

a. Motion ☐ b. Sound ☐ c. Looks ☐

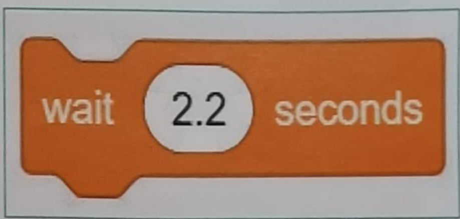
5 Choose the correct option to run the code.

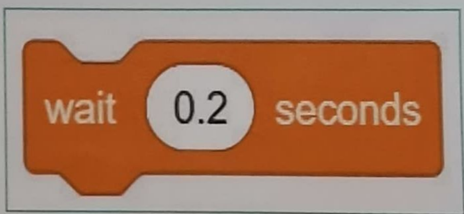
a. ☐ 

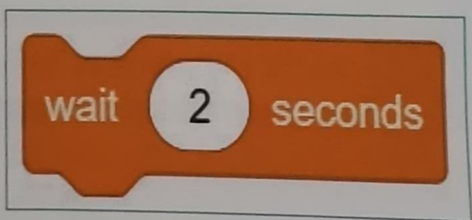
b. ☐ 

c. ☐ 

6 Select the correct block to wait for 2 seconds.

a. ____________ 

b. ____________ 

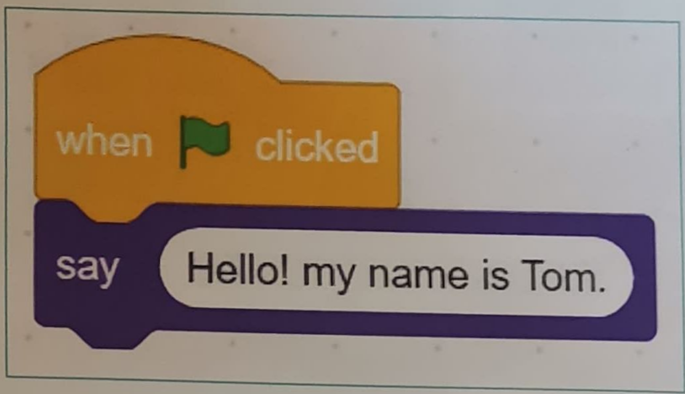
c. ____________ 

Exercise

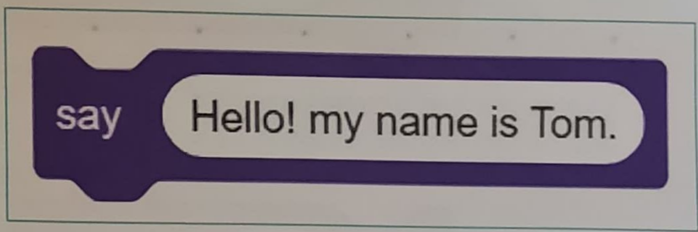
Tick Mark ☑ the correct answer

3. Which block can be used to show the dialogue, “Hello! My name is Tom.”?

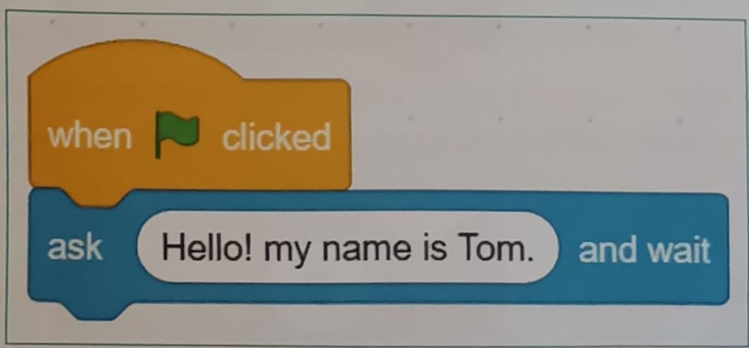
a. ____________



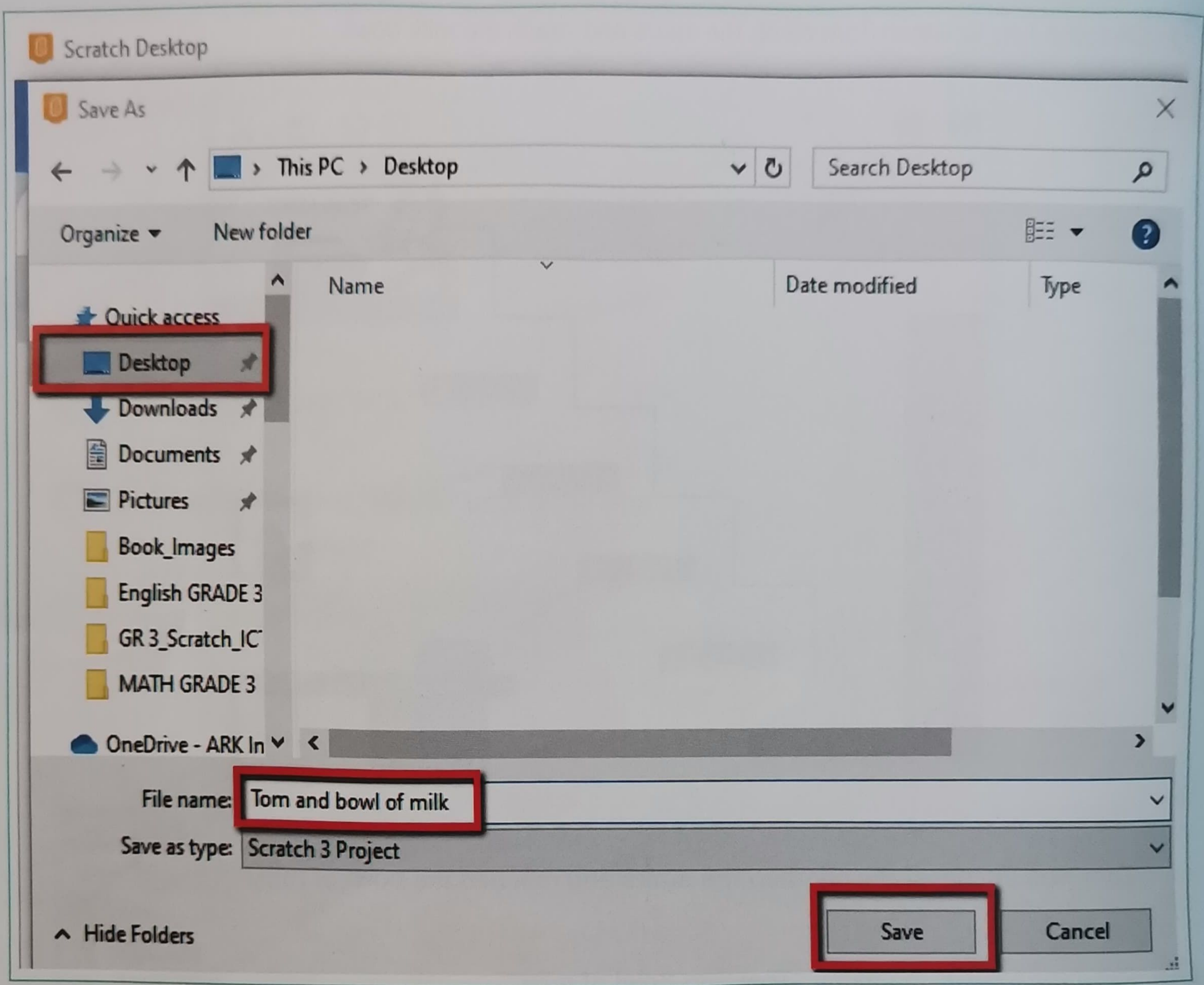
b. ____________



c. ____________



Activity

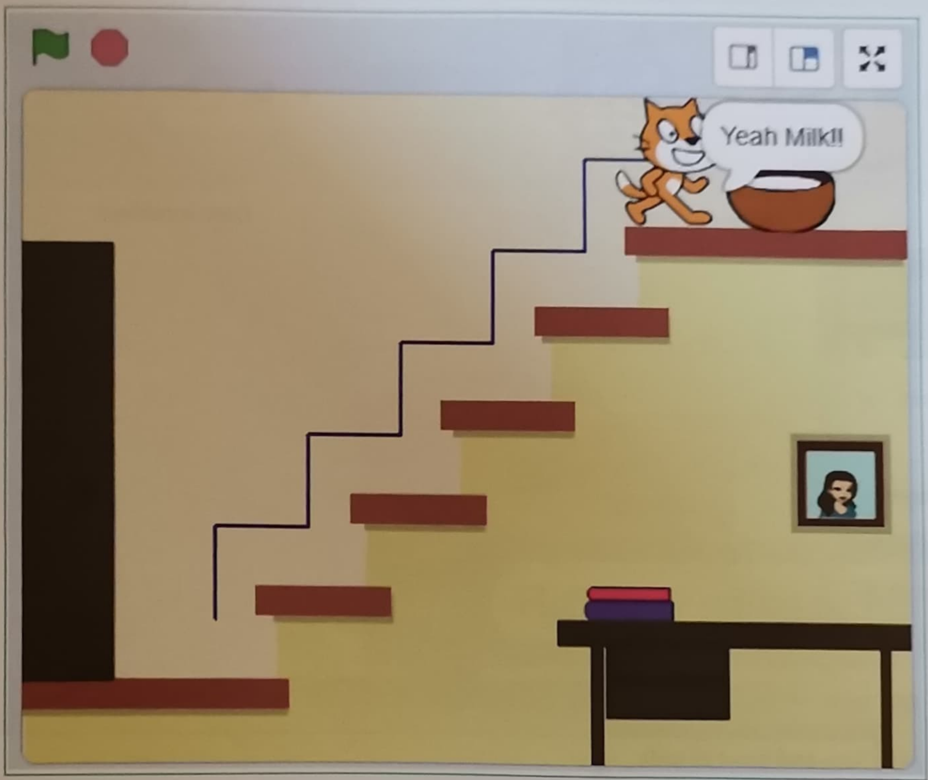


In this lesson, Tom has successfully climbed the stairs using **motion**, **repeat** and **wait** blocks to reach the full bowl of milk. Tom also expressed his happiness on reaching the milk bowl. For this, we used the speech and sound blocks.

ICT360

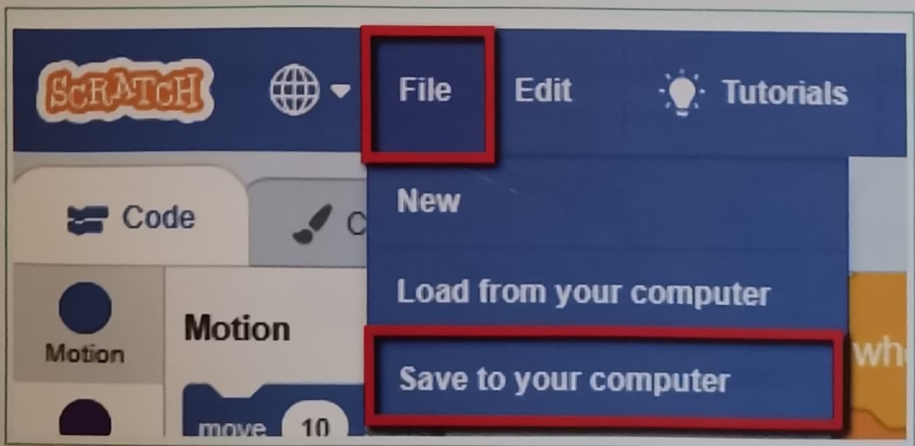
Activity

12 Click the flag to watch Tom climb the stairs and reach the milk bowl.



Tom has successfully climbed the stairs and reached the bowl of milk.

13 Save the file with the name **"Tom and bowl of milk"**.



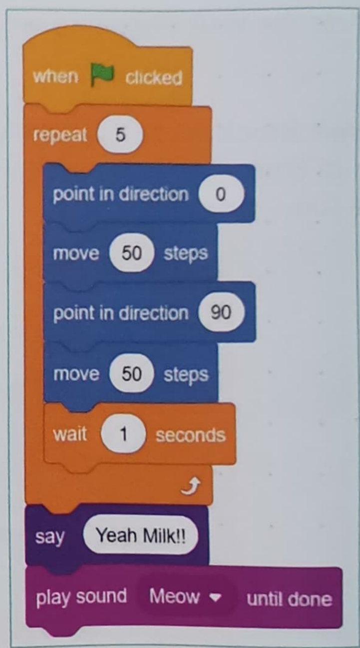
Activity

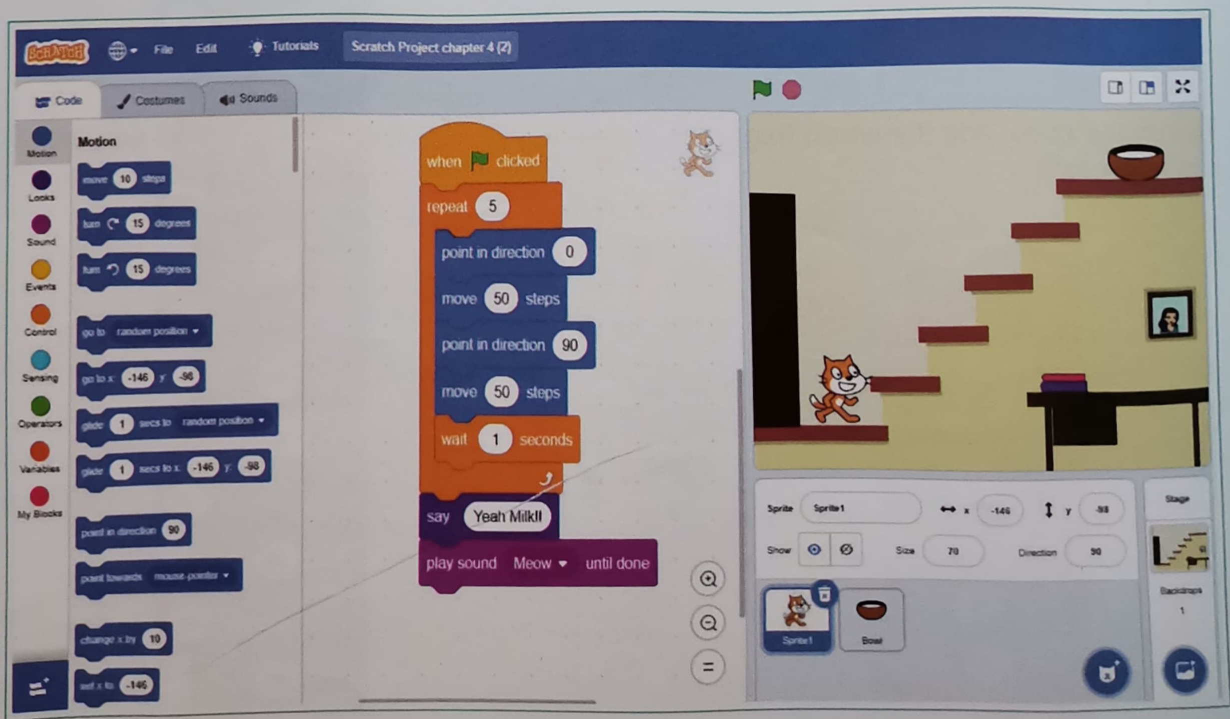
When Tom reaches the milk bowl, he wants to say that he is happy.

We will add a **speech bubble** after Tom reaches the milk bowl and **sound** for Tom to talk. Follow these steps:

10. Add the **Say "Yeah! Milk"** block.

11. Add the **Play sound "Meow" until done** block.





Final stage

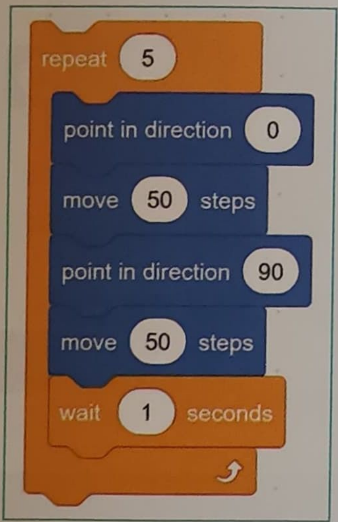
Activity

The repeat block makes it easy and quick. to repeat steps

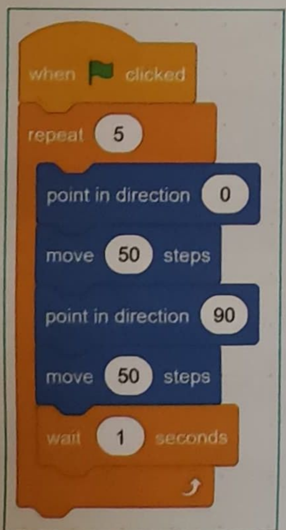
Now,

8 Add the **wait** block and set it to **1 second.**

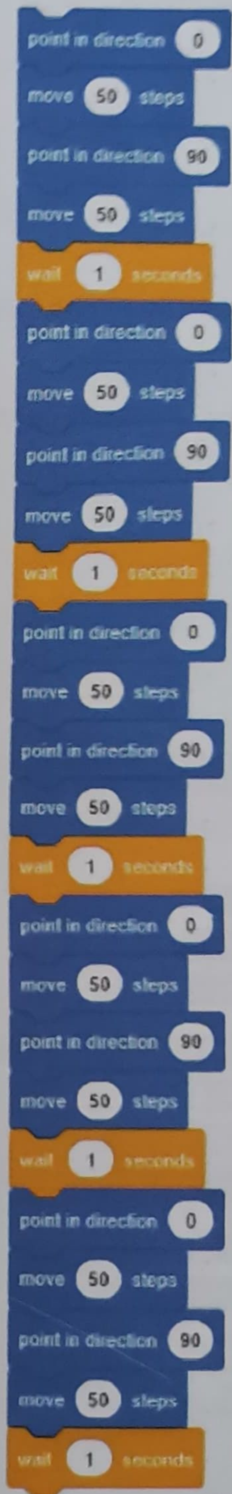
The **Wait Block** is used to wait for a specific period of time. It will add a delay before the next step is run.



9 To run the code, add the **when flag clicked** block.

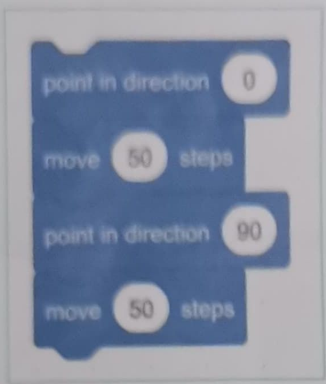


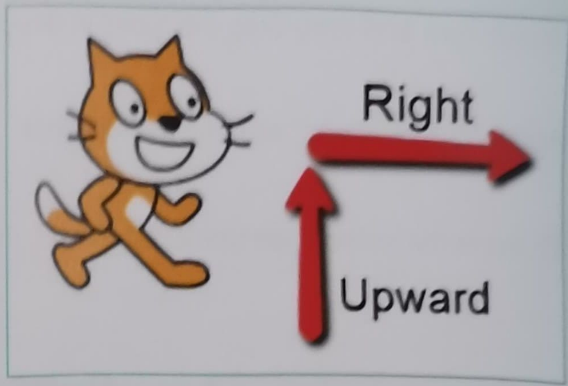
If we don't use the repeat block, we will have to add the same steps five times like this:



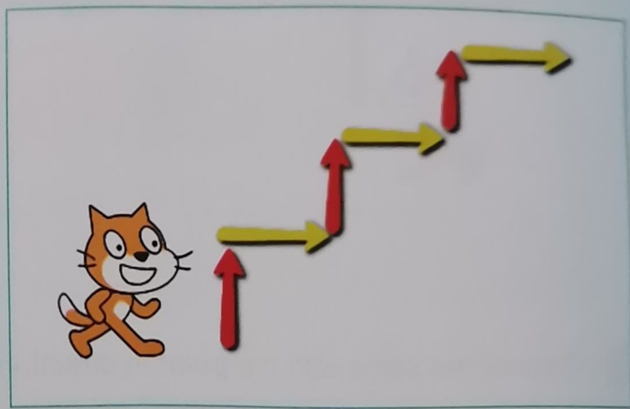
Activity

5. These four blocks put together will move Tom upwards and towards the right.



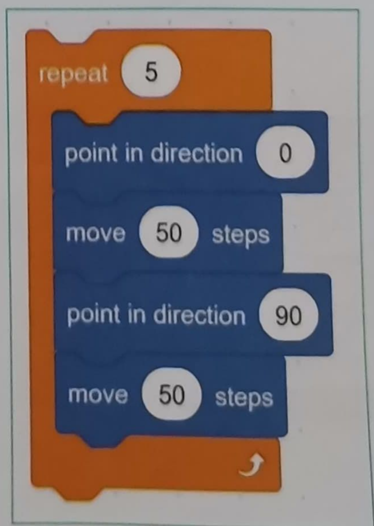


6. To make Tom climb all the stairs, add the **repeat** block.



The **Repeat Block** is used as a loop. It repeats the same instructions until the specified output is reached.

7. To move Tom upwards, we need to repeat this motion five times for five steps.



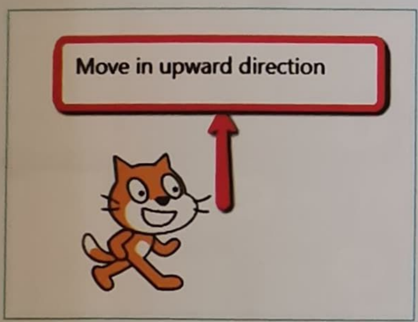
Activity

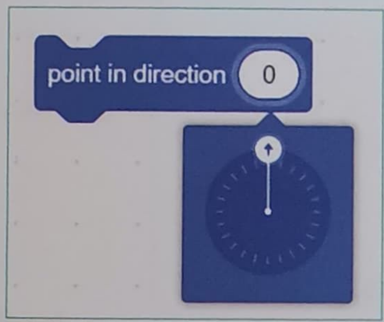
Now, we will give instructions to Tom for him to reach the milk bowl

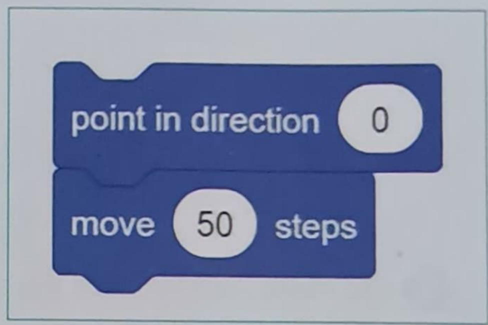
1. Select Tom and drag-drop the "point in direction 0" block.

2. Add "Move 50 steps" with the Motion block.

To move the sprite in an upward direction, the **point in direction** and **move block** work together.

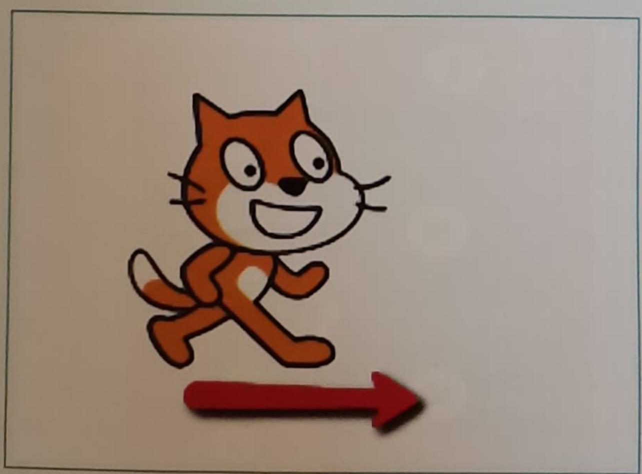


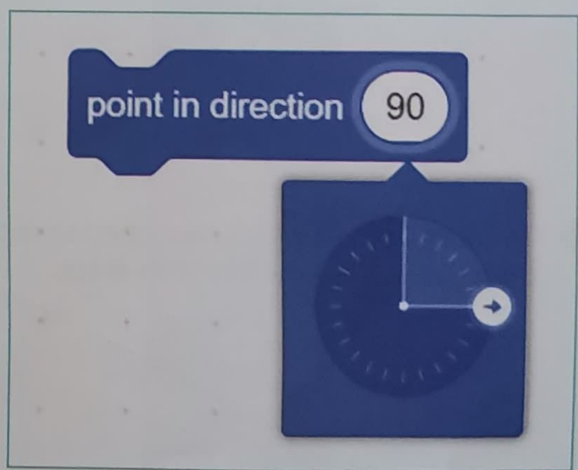




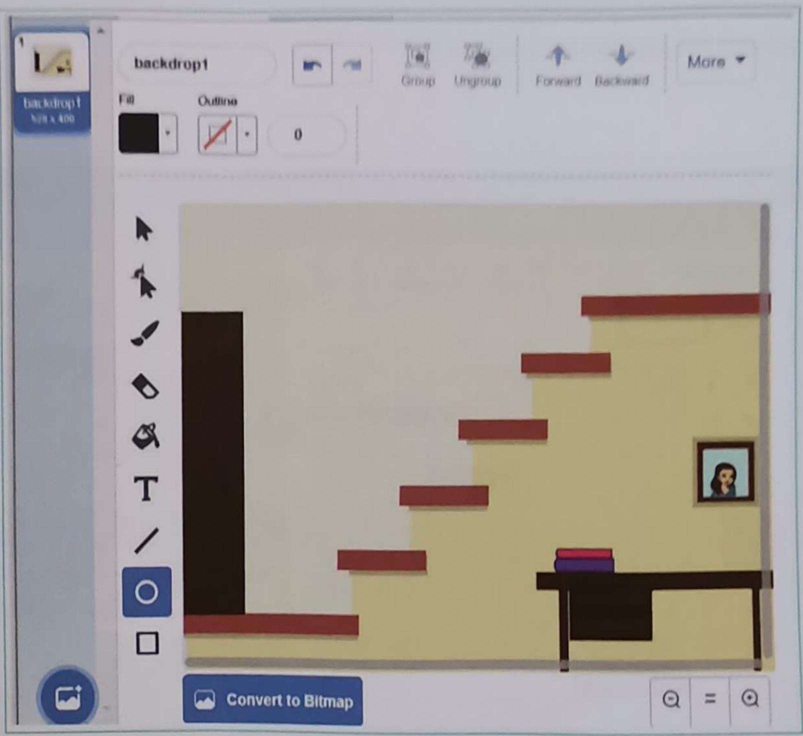
3. Repeat the same with the **point in direction 90** and move **50 steps** blocks.

4. This will move Tom towards the right.





Activity

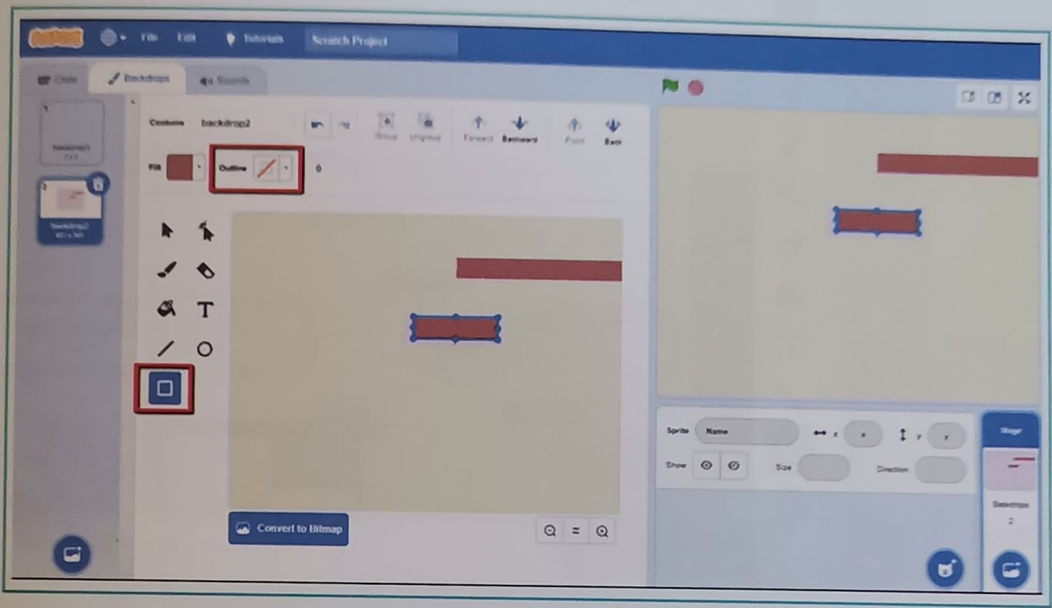


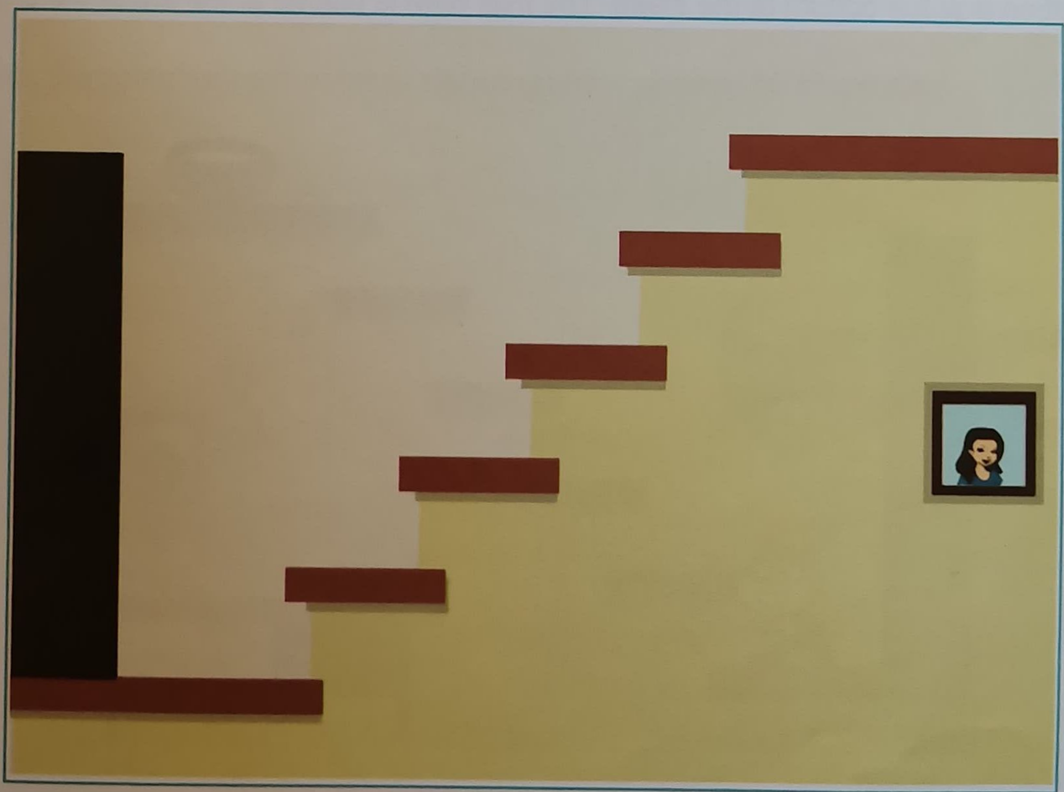
Tom's start position is at the bottom of the stairs. The milk bowl is placed at the top of the staircase.

Start positions of Tom and the milk bowl

Activity

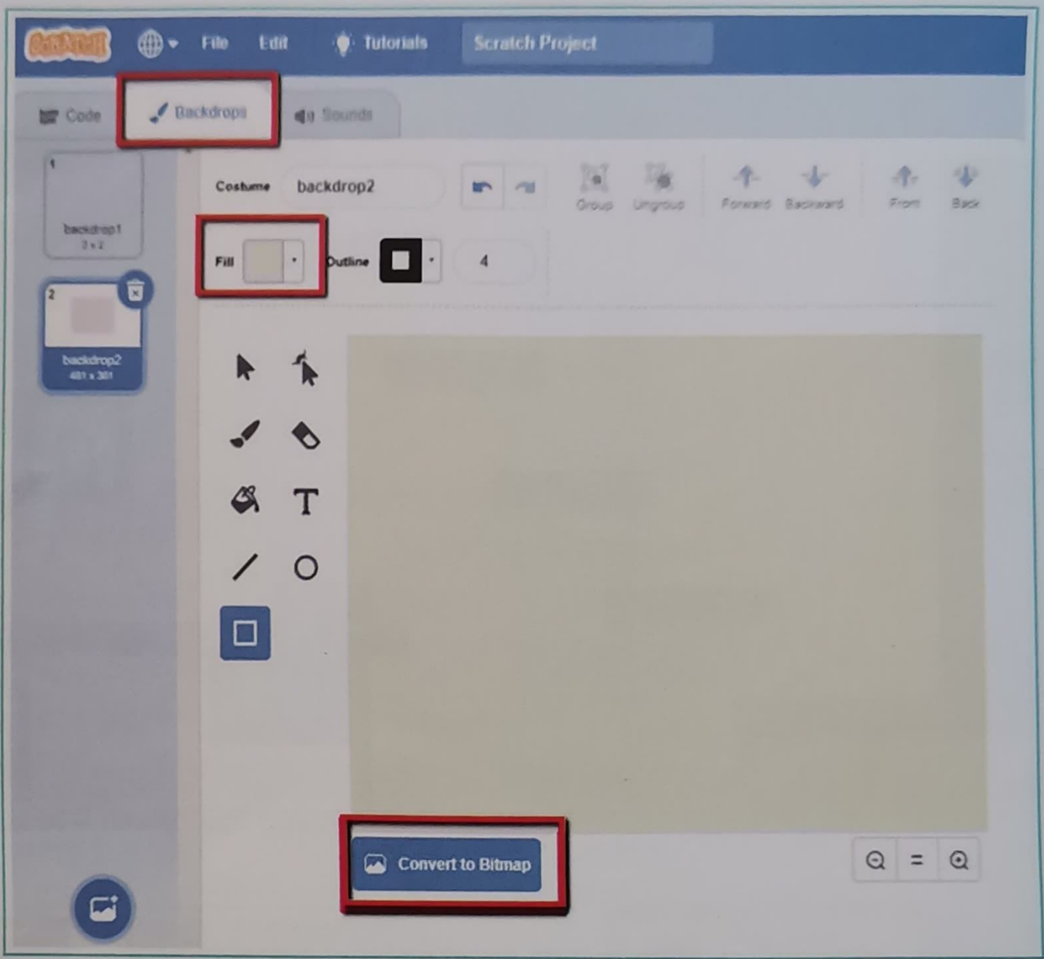
4 Here are some steps to create the backdrop:





Activity

2 Select the colour, use the **fill tool** and click on the black backdrop (fill the colour in vector mode and then come to bitmap).

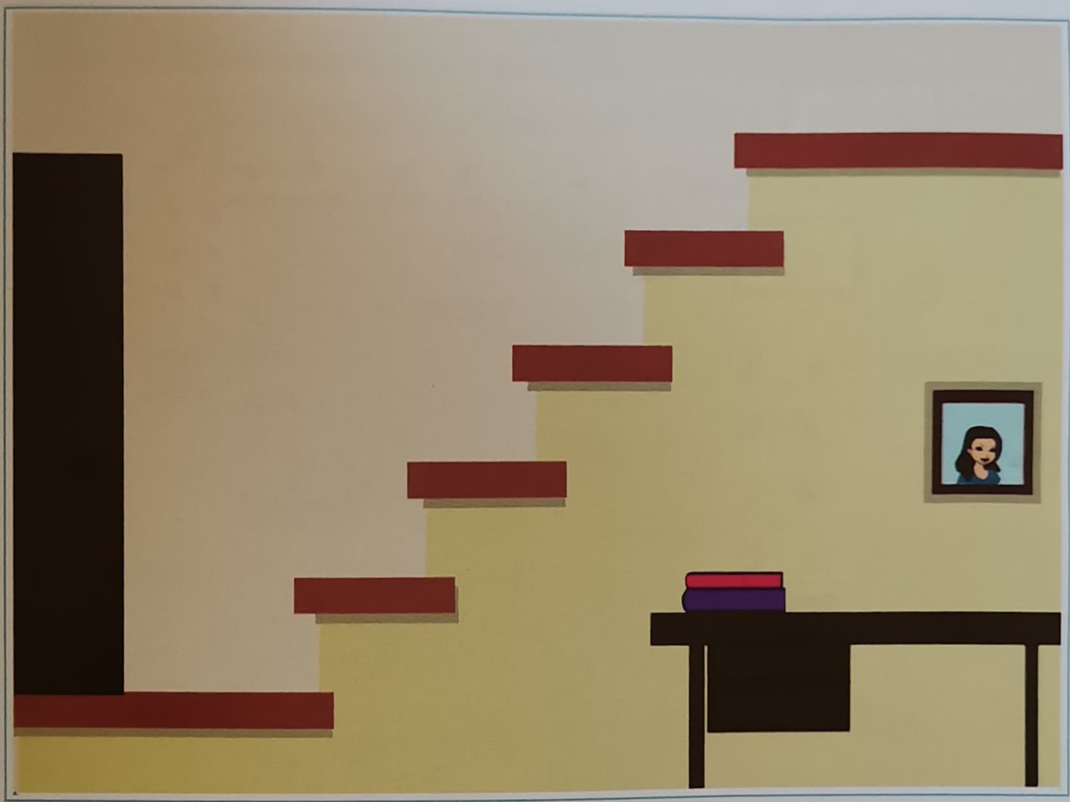


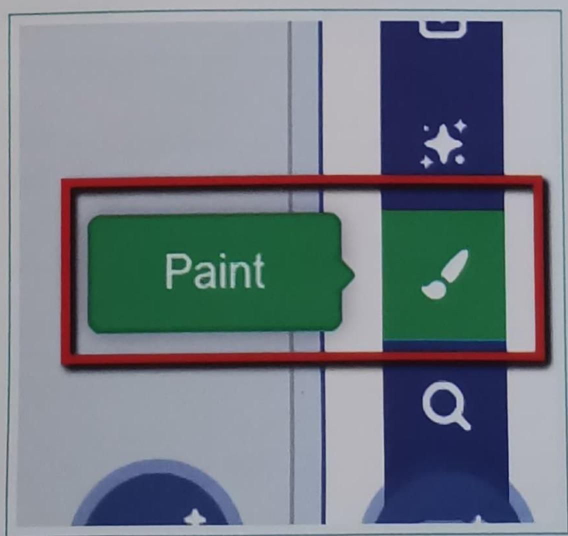
3 To draw the stairs, use the rectangle tool. You can add extra elements like a door and a table.

Activity

To create the stairs on the stage

Select a backdrop from the backdrop library or create a backdrop as shown below:

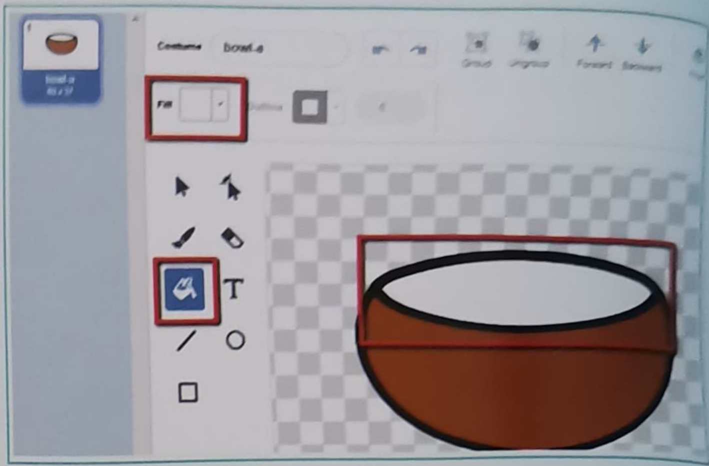




1. Select **Paint** and draw a staircase backdrop with the rectangle, line, and circle tools.

Activity

In the previous lesson, the milk bowl did not have enough milk in it. But now, we will fill the bowl up to the brim.



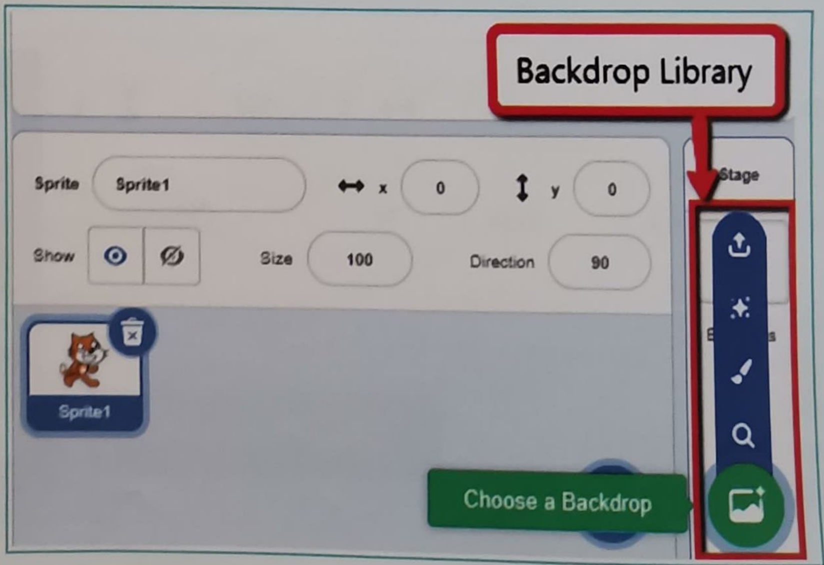
2. To show milk in the bowl, fill it with white colour. You can also use it as it is.

Add a Backdrop

We can select a backdrop from the **Backdrop Library.**

There are four options to select a backdrop:

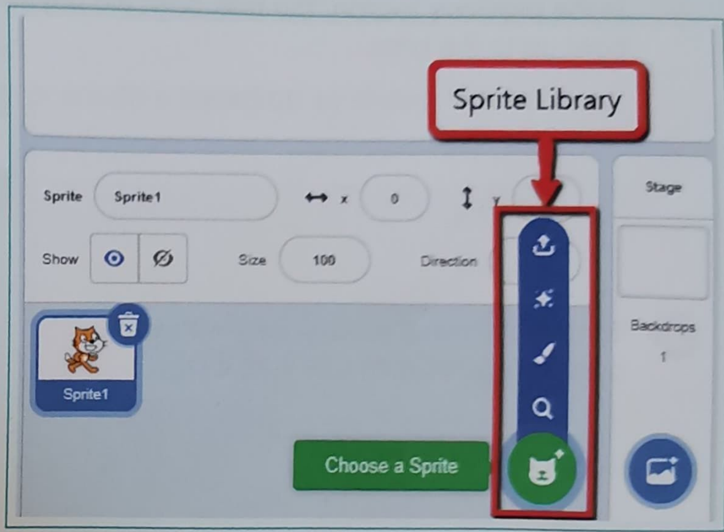
1. **Upload backdrop:** Select a backdrop from the computer
2. **Surprise:** Generate a random backdrop
3. **Paint:** Draw a new backdrop or make changes to the given backdrop
4. **Choose a backdrop:** Select a backdrop from the Backdrop library



Activity

Add a Sprite (TOM) to the Stage

Scratch has many different sprites.
They are stored in the Sprite Library.

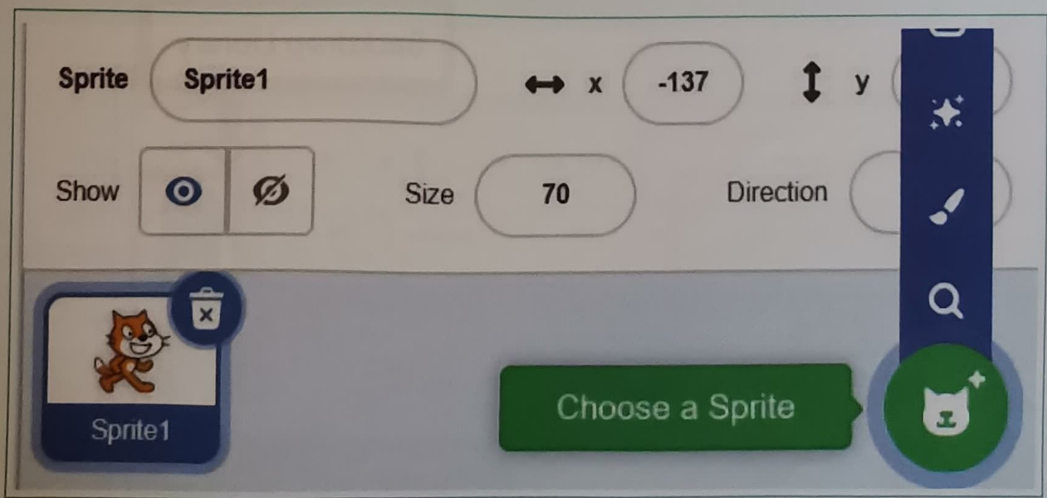


There are four options to select a sprite:

1. **Upload Sprite:** Select a sprite from the computer
2. **Surprise:** Generate a random sprite
3. **Paint:** Draw a new sprite or make changes to the given sprite
4. **Choose a Sprite:** Select a sprite from the Scratch library

Add a Sprite (BOWL) to the Stage

1. Select **Paint** and draw a staircase backdrop with the rectangle, line, and circle tools.



Exercise

Tick Mark ☑ the correct answer

1. Which of the following does not belong to Scratch?

   a. Pencil Tool ☐ b. Sprite ☐ c. Code ☐

2. Which library allows us to choose a backdrop?

   a. Sprite library ☐ b. Code palette ☐ c. Backdrop library ☐

Activity

Earlier, Tom crossed a farm to reach the bowl of milk. But now, he has to climb up a flight of stairs to reach the milk bowl which is at the top.

Draw the path which Tom will travel along to reach the milk bowl.

Recall:

The cat is the default sprite in Scratch.

4. Loops

Loop

A loop is formed when a sequence runs continuously. A Loop does not stop until a specific output is achieved.

For example: We eat food one bite at a time. We dip our spoon into the bowl of food and then put the spoon in our mouth. Then, we chew our food. We repeat the same loop until our hunger is satisfied.

Fill up the spoon with food.

Bring the spoon up to your mouth.

Put the food into your mouth.

Chew your food completely.

Swallow your food.

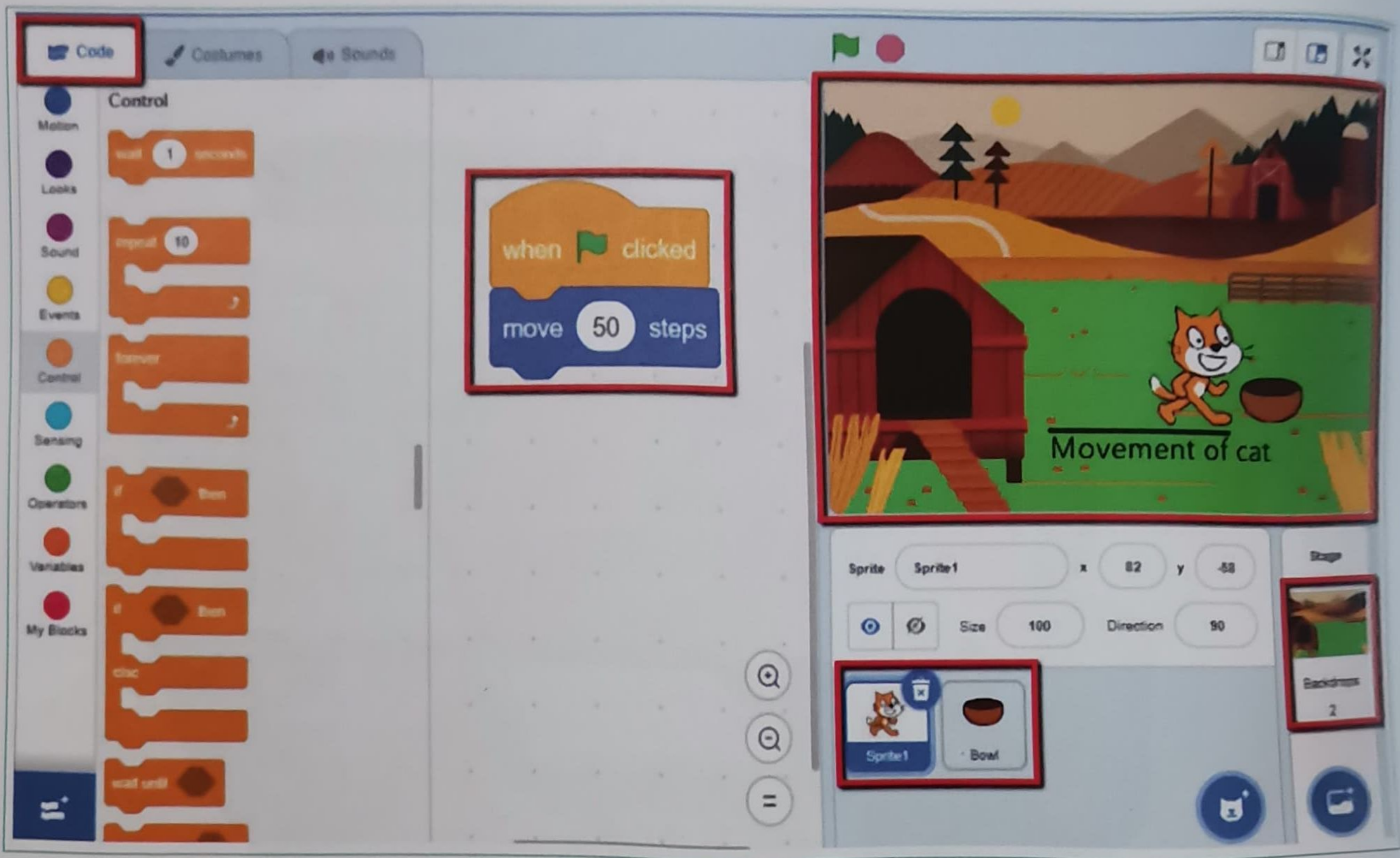
Repeat this loop until your hunger is satisfied.

Loops are also used in machines.

For example: When we turn on the water wipers in a car, they keep moving in a loop. They only stop when we press a stop button.

In Scratch, to make any animation or game, we use the concept of structures. We combine elements such as sprites, backdrops, and codes to create structures.

Example: We created a code for Tom to reach the bowl of milk.



Bytes

Animation is a way to add movement to drawings, models, and non-living objects. This makes them appear to move as if they are alive. All our favourite cartoon shows and movies, like 'Toy Story', are animated. We can animate objects and drawings on a computer with animation software.

Bytes

Computer games are programs that allow a player to play or interact with imaginary or virtual characters. For example, video games, car racing games, and snakes and ladders.

Loops

04

Structure:

A Structure is made when many parts are put together. A skyscraper, a human body, and even a sentence are all examples of a structure.

For example:

Wooden sticks can be stacked on top of each other to make the structure look like a Christmas tree.